

ADC Fundamentals & ADC Errors

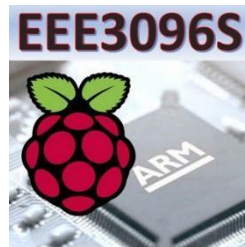
Sampling & Data Conversion: Module 4, Lecture 2 of 4

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What Is an ADC?

A divider, not a mystery box



ADC · THE ONE THING

An ADC does not measure a voltage. It reports **what fraction of** V_{REF} the input is, as one of 2^n whole numbers. It is a **divider** with a rounded-off answer.

mixed-signal

analog in, digital out

ratiometric

everything is relative to V_{REF}

quantised

a code is a band, not a point



The Transfer Function: One Formula for Everything

$$\text{code} = 2^n \times G \times \frac{A_{IN}}{V_{REF}}$$

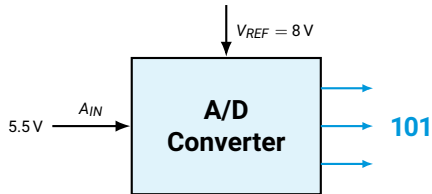
Symbol	Meaning	Note
n	number of output bits (resolution)	sets how many codes exist
G	gain factor	usually exactly 1 , but check the datasheet
A_{IN}	analog input voltage (or current)	the thing you are measuring
V_{REF}	reference voltage (or current)	what the input is compared <i>against</i>

“The output is 2^n times the fraction A_{IN}/V_{REF} .” Every ADC number you ever read is that fraction in disguise, which is why V_{REF} errors become *signal* errors.



A 3-Bit Converter, Concretely

3 bits $\rightarrow 2^3 = 8$ **codes**. With $V_{REF} = 8\text{ V}$, one LSB is $8/8 = 1\text{ V}$.

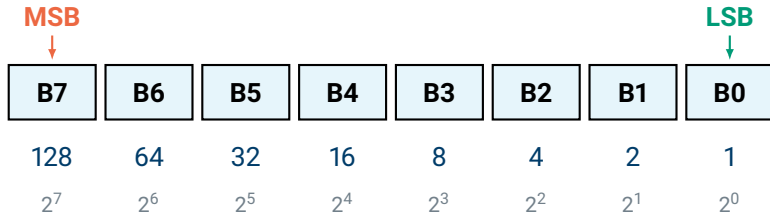


Code	Input band
000	$0\text{ V} \leq A_{IN} < 1\text{ V}$
001	$1\text{ V} \leq A_{IN} < 2\text{ V}$
010	$2\text{ V} \leq A_{IN} < 3\text{ V}$
011	$3\text{ V} \leq A_{IN} < 4\text{ V}$
100	$4\text{ V} \leq A_{IN} < 5\text{ V}$
101	$5\text{ V} \leq A_{IN} < 6\text{ V}$
110	$6\text{ V} \leq A_{IN} < 7\text{ V}$
111	$7\text{ V} \leq A_{IN} < 8\text{ V}$

$2^3 \times 1 \times \frac{5.5}{8} = 5.5 \rightarrow \text{truncate} \rightarrow \mathbf{5 = 101}$. Note what the answer does *not* tell you: anything between 5 V and 6 V gives the same code.



MSB and LSB: Every Bit Has a Weight



- 1 The **MSB** carries weight 2^{n-1} : half of full scale, on its own.
- 2 The **LSB** carries weight 1: it is the smallest change the converter can represent.
- 3 “One LSB” is used two ways: the *bit*, and the *voltage* that bit is worth. Context tells you which.



One LSB = $V_{REF}/2^n$: the Number That Matters

The LSB voltage depends on **two** things only: the reference, and the resolution.

V_{REF}	Bits	1 LSB	V_{REF}	Bits	1 LSB
1.00 V	8	3.9062 mV	2.048 V	10	2.0000 mV
1.00 V	12	244.14 μ V	2.048 V	12	500.00 μ V
2.00 V	8	7.8125 mV	4.00 V	8	15.625 mV
2.00 V	10	1.9531 mV	4.00 V	10	3.9062 mV
2.00 V	12	488.28 μ V	4.00 V	12	976.56 μ V

Two ways to a smaller LSB

Add bits, **or** lower V_{REF} . Note 2.048 V is chosen so the LSB lands on a round number.

Neither is free

More bits costs money. A smaller V_{REF} costs **input range**, and both push the signal closer to the noise floor.

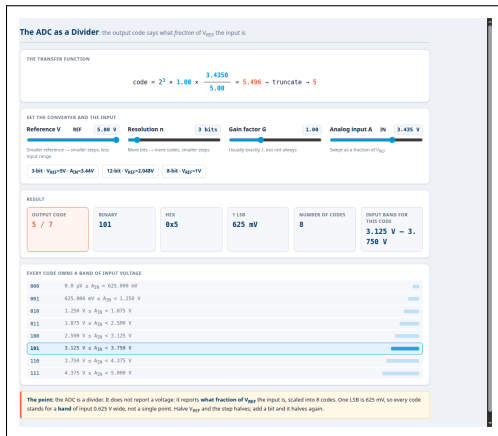


Live Demo: Turn the Divider Yourself

LIVE DEMO

The ADC as a Divider

Set V_{REF} , resolution, gain and input, and watch the formula, the LSB size and the code band update together.



Reproduce the 3-bit, 8 V example, then switch to 12-bit with $V_{REF} = 2.048$ V and confirm the 500 μ V LSB from the table. Ask what happened to the input range.

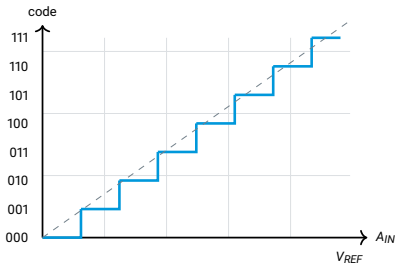
→ `interactive/adc-divider-model.html`

Quantisation Error

The error you cannot design away



Quantisation Error: 0 to 1 LSB



Only a **smaller LSB** (more bits, or a smaller V_{REF}) shrinks it.

1

Each code covers a **band** of input, a *quanta*, q . The converter cannot tell you where inside that band the input actually was.

2

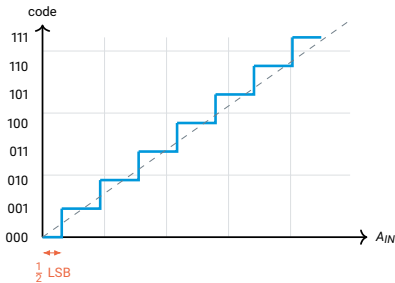
Truncating, the error climbs from **0 to 1 LSB** then snaps back: a **sawtooth**, once per code.

3

This is **quantisation uncertainty**. It is not a fault, not noise, and not fixable: it follows from having finitely many codes.



Adding $\frac{1}{2}$ LSB Offset: Re-centre, Don't Remove



1 Shift every transition **down by $\frac{1}{2}$ LSB**: the first code change now happens at $\frac{1}{2}$ LSB, not 1 LSB.

2 The error becomes $\pm \frac{1}{2}$ **LSB**, symmetric about zero instead of always positive.

3 The **span is still exactly 1 LSB**. Nothing was removed. The bias was removed.

This is why the bound is quoted as $\pm \frac{1}{2}$ LSB. Datasheets differ on which convention they use, so check.



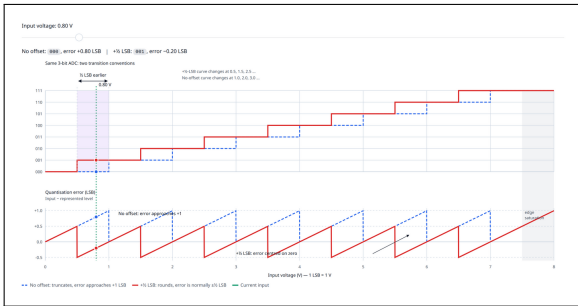
Live Demo: Watch the Sawtooth Move

LIVE DEMO

Quantisation Error & the $\frac{1}{2}$ LSB Offset

One input slider drives a 3-bit ADC both ways at once: no offset and $+\frac{1}{2}$ LSB, overlaid.

Drag the slider: the no-offset error climbs toward +1 LSB, the $+\frac{1}{2}$ LSB error stays centred on zero. Same input, same comparator, different convention.
→ interactive.adc-offset-comparison.html



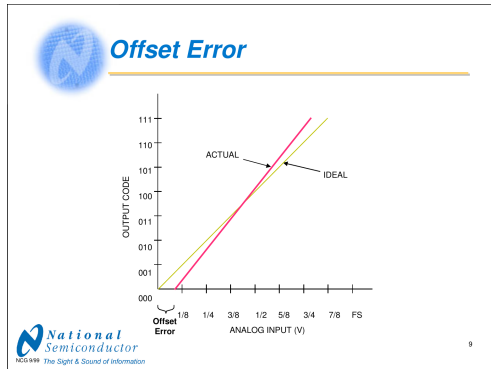
SEGMENT 3 OF 4

Static Errors

Everything you can see on the transfer curve



Offset Error: the Whole Curve Shifts



1

Ideally, an input of $q/2$ is what *just* triggers the first code transition. Any deviation from that is **offset error**.

2

Also called **zero-scale error**. It is **constant**: every code moves by the same amount.

3

The slope is untouched, and the step size is untouched, which is why offset is the **easiest error to calibrate out**: subtract one number.

Quoted in LSB, in volts, or as % of full scale.

Figure: N. Gray, "ABCs of ADCs," National Semiconductor, Rev. 3, June 2006, p. 9. © National Semiconductor Corporation.



Full-Scale Error: Where the Last Transition Lands

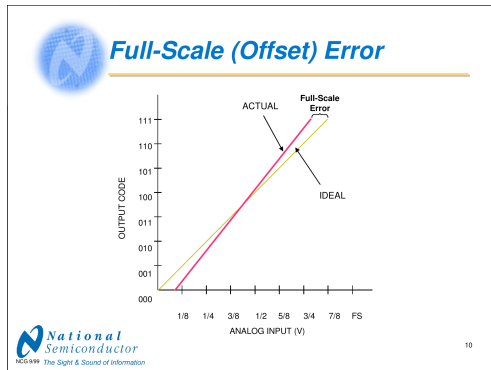


Figure: N. Gray, "ABCs of ADCs," National Semiconductor, Rev. 3, June 2006, p. 10. © National Semiconductor Corporation.

1

The error in the input voltage that causes the **last** transition, into the top code.

2

It contains **both** effects at once: part of it is offset, and the rest is a wrong slope.

3

Sometimes called **full-scale offset error**. Quoted in LSB, volts, or % of ideal full scale.

Subtract the offset from full-scale error and what is left is pure slope. That leftover has its own name: next slide.



Gain Error: the Slope Is Wrong

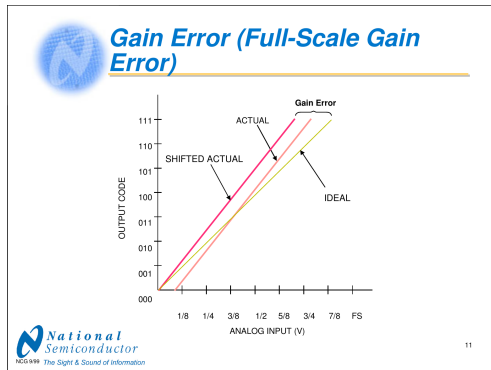


Figure: N. Gray, "ABCs of ADCs," National Semiconductor, Rev. 3, June 2006, p. 11. © National Semiconductor Corporation.

1

Gain error = full-scale error with the offset error **subtracted out**. Shift the curve so zero-scale error is zero; what remains is gain.

2

It is a **deviation in slope**, so the deviation **grows with input**: near zero at the bottom, worst at full scale.

3

That growth-with-input signature is exactly what distinguishes it from offset, which is flat everywhere.

Usually quoted in %FSR or in LSB.

Linearity: Two Different Questions

DNL: Differential Non-Linearity

Also written **DLE** (differential linearity error).

Asks: **is every step exactly 1 LSB wide?**

A **small-scale**, code-to-code error.

INL: Integral Non-Linearity

Also written **ILE** (integral linearity error).

Asks: **is the overall curve straight?**

A **large-scale**, whole-transfer-function error.

1 The word to hang onto is **“differential”**: DNL is a **difference** between the ideal and actual width of one code.

2 INL is the **bow**, the maximum departure of the curve from a straight reference line.

Both exclude offset and gain: those are removed first. DNL and INL are what is left after the easy errors have been calibrated away.



DNL: Is Every Step 1 LSB Wide?

3-bit example, $V_{REF} = 2.0\text{ V}$ so 1 LSB = 250 mV. Measure the width of each code.

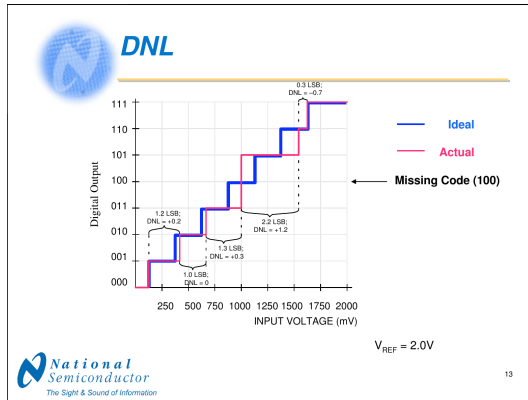


Figure: N. Gray, "ABCs of ADCs," National Semiconductor, Rev. 3, June 2006, p. 13. © National Semiconductor Corporation.

$$\text{DNL} = (\text{actual code width}) - 1 \text{ LSB.}$$

Missing Codes: When a Band Closes Completely

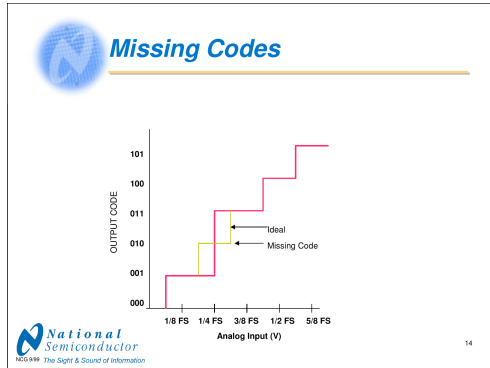


Figure: N. Gray, "ABCs of ADCs," National Semiconductor, Rev. 3, June 2006, p. 14. © National Semiconductor Corporation.

1 If a code's band closes to **zero width**, then **no input voltage anywhere** produces it. The code is **missing**.

2 That happens exactly when **DNL = -1.0 LSB**: the width has shrunk by a whole LSB, to nothing.

3 So the guarantee to look for is "**no missing codes**", which is the same statement as $DNL > -1$.

Critical in servo and control loops: a missing code is a step the loop can never settle on, so it hunts.

INL: Is the Curve Straight?

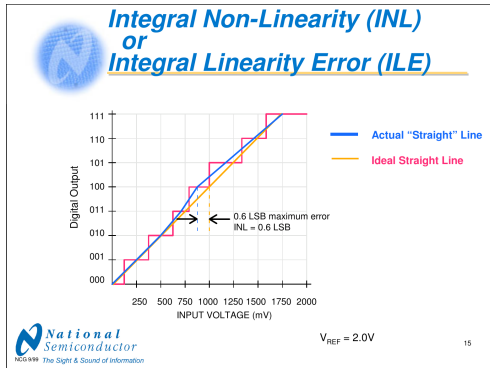


Figure: N. Gray, "ABCs of ADCs," National Semiconductor, Rev. 3, June 2006, p. 15. © National Semiconductor Corporation.

1

INL is the **maximum deviation** of the transfer curve from a straight reference line, once offset and gain are removed.

2

It is **not** the sum of DNLs, but it is caused by them: the size and distribution of the DNL errors determine the bow.

Live Demo: Dial In One Error at a Time

LIVE DEMO

Static ADC Errors

Offset, gain, code-width spread and bow as four independent sliders on one transfer curve, with DNL, INL, missing codes and TUE computed live.

Push code-width spread until a DNL bar hits -1.00 and watch a code vanish from the staircase and appear in the “missing codes” box. That is the “no missing codes” spec, demonstrated.

→ [interactive/adc-static-errors.html](https://interactive.adc-static-errors.html)



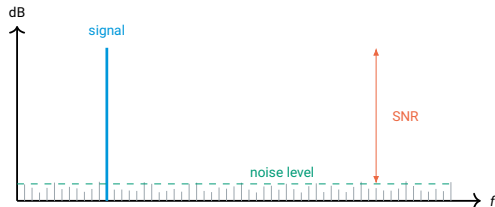
SEGMENT 4 OF 4

Dynamic Performance

What happens once the signal moves



SNR: Signal Against the Noise Floor



For an **ideal** converter, the only noise is quantisation noise:

$$\text{SNR} = 6.02n + 1.76 \text{ dB}$$

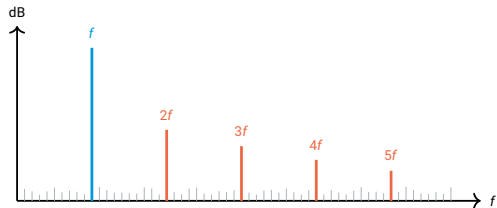
Noise in an ADC comes from four places:

- 1 **Quantisation noise:** the $\pm \frac{1}{2}$ LSB floor, set purely by resolution.
- 2 **The converter itself:** thermal and device noise on-chip.
- 3 **The application circuit:** layout, supplies, grounding. Yours to get wrong.
- 4 **Jitter:** clock uncertainty, and it worsens with input frequency.

SNR **excludes** harmonics, and degrades as input frequency rises.



THD: Harmonics the Input Never Had



$$\text{THD} = \frac{\sqrt{V_{f2}^2 + V_{f3}^2 + \dots + V_{fn}^2}}{V_{f1}}$$

1

Put a **spectrally pure sine** in. A perfectly linear converter gives you back **one** frequency component.

2

Any non-linearity in the transfer function **manufactures harmonics** at $2f, 3f, 4f \dots$ that were never in the input.

3

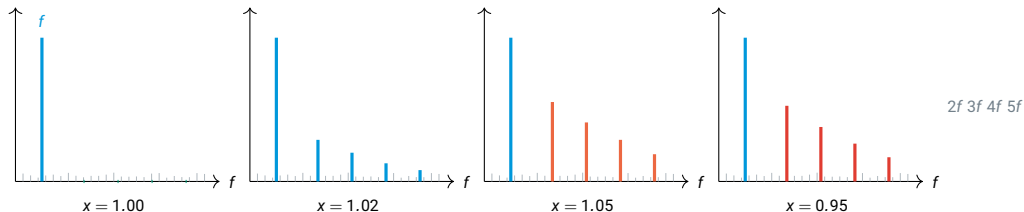
THD is the ratio of the RMS sum of those harmonics to the fundamental. Expressed as a percentage or in dB.

Symmetric non-linearity $\rightarrow$ odd harmonics.
Half-wave effects $\rightarrow$ even harmonics. THD worsens with amplitude and with frequency.



Where the Harmonics Come From

Model the real transfer function as $V_{out} = (V_{in})^x$ and sweep x . The *curve* barely moves; the *spectrum* does not stay quiet.



SINAD: Noise and Distortion Together

$$\text{SINAD} = -20 \log_{10} \sqrt{10^{\frac{-\text{SNR}}{10}} + 10^{\frac{\text{THD}}{10}}}$$

1

SNR counts the noise. THD counts the harmonics. **SINAD counts both**, weighted equally.

2

It is the RMS of the signal against **everything else** below $f_s/2$, harmonics included, d.c. excluded.

3

Comparing the signal against everything unwanted at once makes SINAD the **single best overall measure** of dynamic performance.

4

You would expect it to peak where SNR and $-\text{THD}$ are equal, but on modern parts distortion is so low that SNR dominates.



ENOB: The Bits You Actually Get

An **ENOB of 7.0** means the part performs, as far as SINAD is concerned, exactly like a **perfect 7-bit converter**.

$$\text{ENOB} = \frac{\text{SINAD} - 1.76}{6.02}$$

1

It comes straight from inverting $\text{SNR} = 6.02n + 1.76$: substitute the *measured* SINAD and solve for n .

3

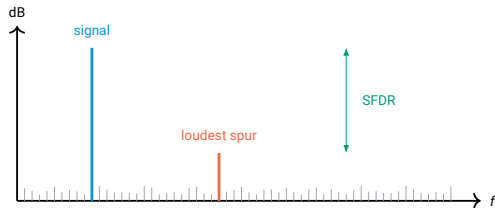
ENOB **degrades** as frequency rises and as input level falls, for the same reasons SNR and THD do.

2

So ENOB is not new information: it is **SINAD restated in bits**, which is a far more intuitive unit.



SFDR: Distance to the Loudest Spur



1

SFDR is the distance in dB from the fundamental down to the **single largest unwanted component** anywhere in the spectrum.

2

That spur may be a harmonic, or it may not: SFDR does not care what caused it, only how tall it is.

3

It is the spec that matters when you must **see a small signal next to a large one**: comms, radar, spectrum analysis.

It is a *range*, so quote it as a **positive** dB number, even though the spur sits below the fundamental.



Input Dynamic Range (and What It Is Not)

Resolution (bits)	Dynamic range (dB)
6	36.0
8	48.1
10	60.2
12	72.2
14	84.3
16	96.3
18	108.4
20	120.4

$$DR = 20 \log_{10}(2^n - 1) \text{ dB}$$

1

Ratio of the **largest** to the **smallest** resolvable signal: largest code $2^n - 1$, smallest non-zero code 1.

2

A property of **resolution alone**: it says nothing about noise, distortion or spurs.

Do not confuse dynamic range with SFDR. DR is what the code space can span; SFDR is how far down the worst spur sits in a real measurement.



Live Demo: Five Specifications, One Spectrum

LIVE DEMO

Dynamic Performance

Drive a simulated converter with adjustable noise and non-linearity, and read SNR, THD, SINAD, ENOB and SFDR off the same FFT.



Add non-linearity until a harmonic overtakes the noise floor: SFDR flips from noise-set to harmonic-set, and ENOB follows SINAD down. Then back off to -20 dBFS and watch SNR fall 1 dB per dB.

→ `interactive/adcdynamic-spectrum.html`



Resolution is a promise. ENOB is the delivery.

An ADC is a divider: the code is $2^n G A_{IN}/V_{REF}$, and every code is a band, not a point.

Static errors (offset, full-scale, gain, DNL, INL) are what you read off the transfer curve.

Dynamic specs (SNR, THD, SINAD, ENOB, SFDR) are what you get once the signal moves.

Next: L7, the types of ADC, and why architecture is a trade-off surface.

Questions?

